Introduction to Analysis I

Lecture notes, Chapters 1–5 and Appendix

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Chapter 1. Real Numbers and Limits of Sequences

Analysis rests on a precise description of the real numbers. We take $\mathbb{R}$ as given by a short list of axioms — the field axioms, the order axioms, and the completeness axiom — and derive everything else from them. (A construction of $\mathbb{R}$ satisfying the axioms is sketched in Appendix A.)

1.1 Field and order axioms

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$ such that both are associative and commutative, $\cdot$ distributes over $+$, there are identities $0 \neq 1$, every a has an additive inverse $-a$, and every $a \neq 0$ has a multiplicative inverse a^{-1}. A field is **ordered** if there is a subset $P \subset F$ (the positive elements) with

(O1) $a, b \in P \Rightarrow a + b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$;

(O3) for each a, exactly one of $a \in P$, $a = 0$, $-a \in P$ holds (trichotomy).

We write $a < b$ if $b - a \in P$.

Proposition 1.1

In an ordered field: (a) $a < b$ and $b < c$ imply $a < c$; (b) $b < c \iff a + b < a + c$; (c) $a > 0$, $b < c$ imply $ab < ac$, while $a < 0$, $b < c$ imply $ab > ac$; (d) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$; (e) $0 < a < b \Rightarrow 0 < 1/b < 1/a$; (f) for $a, b > 0$, $a < b \iff a^2 < b^2$.

Proof. (a) $c - a = (c - b) + (b - a) \in P$ by (O1). (b) $(a + c) - (a + b) = c - b$. (c) $ac - ab = a(c - b)$; use (O2), and for $a < 0$ note $a > 0 \iff -a < 0$ (trichotomy). (d) If $a > 0$, $a^2 \in P$ by (O2); if $a < 0$, $a^2 = (-a)^2 \in P$. (e) $1/a > 0$ since otherwise $1 = a \cdot (1/a) \leq 0$; then multiply $a < b$ by $1/(ab) > 0$. (f) $b^2 - a^2 = (b - a)(b + a)$ and $b + a > 0$. ∎

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields; $\mathbb{C}$ is not: if $P \subset \mathbb{C}$ satisfied (O1)–(O3), then $-1 = i^2 \in P$ and $1 = 1^2 \in P$, contradicting trichotomy. No finite field is ordered either, since $1 + 1 + \cdots + 1 > 0$ forces all these sums to be distinct.

The **absolute value** is $|a| = a$ for $a \geq 0$ and $-a$ for $a < 0$.

Proposition 1.2

$|a| \geq 0$ with equality iff $a = 0$; $|ab| = |a||b|$; $|a + b| \leq |a| + |b|$; and $\big||a| - |b|\big| \leq |a - b|$.

Proof. The first two are case checks. Since $-|a| \leq a \leq |a|$ and likewise for b, $-(|a| + |b|) \leq a + b \leq |a| + |b|$, which is the triangle inequality. Applying it to $a = (a - b) + b$ gives $|a| - |b| \leq |a - b|$, and symmetrically. ∎

1.2 The completeness axiom

A set $S \subset \mathbb{R}$ is **bounded above** if some u (an *upper bound*) satisfies $s \leq u$ for all $s \in S$. A **least upper bound** or **supremum** $\sup S$ is an upper bound that is $\leq$ every upper bound; $\inf S$ is defined dually.

Definition (Completeness axiom)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Equivalently, every nonempty set bounded below has an infimum: if S is bounded below, the set L of its lower bounds is nonempty and bounded above (by any element of S), and $\sup L = \inf S$. The ordered field $\mathbb{Q}$ is *not* complete: $\{r \in \mathbb{Q} : r^2 < 2\}$ has no rational supremum (Proposition 1.5 and the irrationality of $\sqrt{2}$).

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Theorem 1.3 (Archimedean property)

$\mathbb{N}$ is not bounded above. Consequently, for $a > 0$ and any $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$; in particular $1/n \to 0$ in the sense that for every $\epsilon > 0$ some n has $1/n < \epsilon$.

Proof. If $\mathbb{N}$ were bounded above, let $u = \sup \mathbb{N}$. Then $u - 1$ is not an upper bound, so some $n > u - 1$, and $n + 1 \in \mathbb{N}$ exceeds u: contradiction. For the second claim pick $n > b/a$. ■

Theorem 1.4 (Density of the rationals)

If $a < b$ are real, there is $r \in \mathbb{Q}$ with $a < r < b$. The irrationals are dense as well.

Proof. Choose n with $n(b-a) > 1$ and let m be the least integer with $m > na$ (it exists by the Archimedean property and well-ordering). Then $m - 1 \le na$, so $na < m \le na + 1 < nb$ and $r = m/n$ works. For irrationals apply this to $a/\sqrt{2} < b/\sqrt{2}$ and multiply by $\sqrt{2}$. ■

Proposition 1.5

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness follows from Proposition 1.1(f). Let $S = \{x > 0 : x^2 < a\}$, nonempty (it contains $\min(1, a)/2$) and bounded above by $\max(1, a)$; put $b = \sup S$. If $b^2 < a$, then for small $0 < h < 1$, $(b+h)^2 \le b^2 + h(2b+1) < a$, contradicting that b is an upper bound. If $b^2 > a$, then for small $h > 0$, $(b-h)^2 \ge b^2 - 2bh > a$, so $b - h$ would be a smaller upper bound. Hence $b^2 = a$. ■

The completeness axiom pins $\mathbb{R}$ down: any two complete ordered fields F, G are isomorphic, i.e. there is a bijection $\phi : F \to G$ preserving sums, products and order (one maps $\mathbb{N} \subset F$ to $\mathbb{N} \subset G$, extends to $\mathbb{Q}$, and then sends x to $\sup\{\phi(r) : r \in \mathbb{Q}, r < x\}$). We also record the arithmetic of suprema: for nonempty bounded S, T, $\sup(S + T) = \sup S + \sup T$, and $\sup(ST) = \sup S \cdot \sup T$ if $S, T \subset (0, \infty)$; here $S + T = \{s + t\}$ and $ST = \{st\}$.

1.3 Limits of sequences

Definition (Convergence)

A sequence (x_n) **converges** to x, written $x_n \to x$ or $\lim x_n = x$, if for every $\epsilon > 0$ there is N such that $n \ge N \Rightarrow |x_n - x| < \epsilon$.

Proposition 1.6

(a) A sequence has at most one limit. (b) A convergent sequence is bounded.

Proof. (a) If $x_n \to x$ and $x_n \to y$ with $x \neq y$, take $\epsilon = |x-y|/2$; for large n, $|x-y| \leq |x-x_n| + |x_n - y| < 2\epsilon$, a contradiction. (b) With $\epsilon = 1$, $|x_n| \leq |x| + 1$ for $n \geq N$, and finitely many earlier terms are bounded. ■

Theorem 1.7 (Limit laws)

If $x_n \to x$ and $y_n \to y$, then $x_n + y_n \to x + y$, $ax_n \to ax$, $x_n y_n \to xy$, and $x_n/y_n \to x/y$ when $y \neq 0$ (and $y_n \neq 0$). If $x_n \leq y_n$ for all large n, then $x \leq y$. If $x_n \leq z_n \leq y_n$ and $x = y$, then $z_n \to x$ (squeeze).

Proof. Products: $|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x| \leq M|y_n - y| + |y||x_n - x|$, with M a bound for $|x_n|$. Quotients: it suffices that $1/y_n \to 1/y$; for large n, $|y_n| > |y|/2$, so $|1/y_n - 1/y| \leq 2|y_n - y|/|y|^2$. Order: if $x > y$, then $\epsilon = (x-y)/2$ gives $x_n > y_n$ for large n. Squeeze: $x - \epsilon < x_n \leq z_n \leq y_n < x + \epsilon$ for large n. ■

Example 1.1

(a) For $|a| < 1$, $a^n \to 0$: write $1/|a| = 1 + h$ with $h > 0$; by Bernoulli's inequality $(1+h)^n \geq 1 + nh$, so $|a|^n \leq 1/(1+nh) \to 0$.
(b) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, the binomial theorem gives $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)} \to 0$. Similarly $a^{1/n} \to 1$ for $a > 0$.
(c) If $x_n \to x$ then the averages $(x_1 + \cdots + x_n)/n \to x$ (Cesàro): split the sum at a large index N.

1.4 Monotone sequences, limsup and liminf

Theorem 1.8 (Monotone convergence)

A bounded monotone sequence of reals converges; an increasing one converges to $\sup_n x_n$.

Proof. Let (x_n) increase and $s = \sup\{x_n\}$ (completeness). Given ϵ, some $x_N > s - \epsilon$, and then $s - \epsilon < x_N \leq x_n \leq s$ for $n \geq N$. ■

In particular a series of nonnegative terms converges iff its partial sums are bounded.

Example 1.2 (The number e)

$a_n = (1 + 1/n)^n$ is increasing and bounded. By the binomial theorem $a_n = \sum_{k=0}^{n} \frac{1}{k!} \prod_{j<k}(1 - \frac{j}{n})$; each term increases with n and there are more terms, so $a_n \uparrow$; and $a_n \leq \sum_{k=0}^{n} \frac{1}{k!} \leq 1 + \sum_{k \geq 1} 2^{1-k} = 3$. Its limit is e; comparing with the partial sums one finds $e = \sum_{k \geq 0} 1/k!$.

For a bounded sequence put $\overline{x}_n = \sup_{k \geq n} x_k$ (decreasing) and $\underline{x}_n = \inf_{k \geq n} x_k$ (increasing), and define

$$\limsup_{n \to \infty} x_n = \lim_n \sup_{k \geq n} x_k = \inf_n \sup_{k \geq n} x_k, \qquad \liminf_{n \to \infty} x_n = \sup_n \inf_{k \geq n} x_k.$$

These exist by Theorem 1.8, and $\liminf x_n \leq \limsup x_n$.

Proposition 1.9 (Characterization of limsup)

For a bounded sequence, $a = \limsup x_n$ iff for every $\epsilon > 0$: (i) $x_n < a + \epsilon$ for all sufficiently large n, and (ii) $x_n > a - \epsilon$ for infinitely many n. Dually for $\liminf$.

Proof. If $a = \inf_n \overline{x}_n$, some $\overline{x}_N < a + \epsilon$, giving (i); and every $\overline{x}_n \geq a$, so for each n some $k \geq n$ has $x_k > a - \epsilon$, giving (ii). Conversely (i) gives $\overline{x}_n \leq a + \epsilon$ for large n and (ii) gives $\overline{x}_n \geq a - \epsilon$ for all n, so $\lim \overline{x}_n = a$. ■

Theorem 1.10

A bounded sequence converges to a iff $\limsup x_n = \liminf x_n = a$.

Proof. If $x_n \to a$, then $a - \epsilon < x_k < a + \epsilon$ for $k \geq N$, so $a - \epsilon \leq \underline{x}_n \leq \overline{x}_n \leq a + \epsilon$ for $n \geq N$. Conversely $\underline{x}_n \leq x_n \leq \overline{x}_n$ and squeeze. ■

Thus "(i) for all large n" versus "(ii) for infinitely many n" is the essential difference between the two halves of the characterization. For unbounded sequences one allows the values $\pm\infty$.

Theorem 1.11 (Nested intervals)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_1 \supset I_2 \supset \cdots$, then $\bigcap_n I_n \neq \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) increases and is bounded by b_1; let $a = \lim a_n = \sup a_n$. For fixed m and $n \geq m$, $a_n \leq b_n \leq b_m$, so $a \leq b_m$; also $a \geq a_m$. Hence $a \in \bigcap I_m$. Two points in all I_n differ by at most $b_n - a_n \to 0$. ■

Theorem 1.8 is in fact equivalent to the completeness axiom (given the other axioms), and so is the nested interval theorem together with the Archimedean property.

1.5 Countable and uncountable sets

Sets A, B have the **same cardinality** if there is a bijection $A \to B$. A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.12

(a) A subset of a countable set is countable. (b) A countable union of countable sets is countable. (c) $\mathbb{N} \times \mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. (a) A subset of $\mathbb{N}$ is finite or enumerated increasingly. (c) $(m, n) \mapsto 2^m 3^n$ injects $\mathbb{N} \times \mathbb{N}$ into $\mathbb{N}$; $\mathbb{Z}$ is enumerated as $0, 1, -1, 2, -2, \ldots$; $\mathbb{Q}$ is the image of $\mathbb{Z} \times \mathbb{N}$ under $(p, q) \mapsto p/q$. (b) If $A_m = \{a_{m1}, a_{m2}, \ldots\}$, then $(m, n) \mapsto a_{mn}$ maps (a subset of) $\mathbb{N} \times \mathbb{N}$ onto $\bigcup A_m$; choose for each element the least preimage. ■

Theorem 1.13 (Cantor)

$\mathbb{R}$ is uncountable. In fact no interval $[a, b]$ with $a < b$ is countable.

Proof. Let $x_1, x_2, \ldots$ be any sequence in $[a, b]$. Divide $[a, b]$ into three closed thirds and choose one, I_1, not containing x_1; divide I_1 into thirds and choose $I_2 \subset I_1$ not containing x_2, and so on. By Theorem 1.11 some $x \in \bigcap I_n$, and $x \neq x_n$ for every n. So no sequence exhausts $[a, b]$. ■

Example 1.3

(a) The irrationals are uncountable (otherwise $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be countable), although both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense.
(b) The set of all subsets of $\mathbb{N}$ is uncountable (Cantor's diagonal argument: no $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ is onto, since $\{n : n \notin f(n)\}$ is missed). Writing reals in $[0, 1)$ in binary shows that $[0, 1)$ has

the cardinality of $\mathcal{P}(\mathbb{N})$.
(c) The **Cantor set** $K = \bigcap K_n$, where $K_0 = [0,1]$ and K_{n+1} removes the open middle third of every interval of K_n, has total removed length $\sum 2^n/3^{n+1} = 1$ but is uncountable: it consists of the numbers whose ternary expansions use only the digits 0 and 2.

Chapter 2. Topology of Euclidean Space

We now work in the coordinate space $\mathbb{R}^n$. The notions of this chapter — open and closed sets, compactness, connectedness — are what make limits and continuity work in several variables.

2.1 Euclidean space

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with componentwise addition and scalar multiplication. The **inner product** and **norm** are

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i y_i, \qquad \|x\| = \sqrt{\langle x, x\rangle},$$

and the distance is $d(x, y) = \|x - y\|$.

Theorem 2.1 (Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $d(x, z) \le d(x, y) + d(y, z)$.

Proof. If $y = 0$ there is nothing to prove. Otherwise $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$ for all real t; the choice $t = \langle x, y\rangle / \|y\|^2$ gives $\langle x, y\rangle^2 \le \|x\|^2\|y\|^2$, with equality iff $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

We also use $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, which lets us pass between the Euclidean norm and coordinates. The **open ball** is $N(x, r) = \{y : \|y - x\| < r\}$.

2.2 Open and closed sets

Definition (Open and closed sets)

$U \subset \mathbb{R}^n$ is **open** if every $x \in U$ has some ball $N(x, r) \subset U$. F is **closed** if its complement is open. A point x is a **limit point** of A if every ball $N(x, r)$ contains a point of A other than x; A' denotes the set of limit points. The **closure** is $\overline{A} = A \cup A'$, the **interior** A° is the set of x with a ball inside A, and the **boundary** is $\overline{A} \setminus A^\circ$. Points of $A \setminus A'$ are **isolated**.

Open balls are open (by the triangle inequality, $N(y, r - \|y - x\|) \subset N(x, r)$).

Proposition 2.2

$\emptyset$ and $\mathbb{R}^n$ are open; arbitrary unions and finite intersections of open sets are open. Dually, arbitrary intersections and finite unions of closed sets are closed.

Proof. For a finite intersection, take the smallest of the finitely many radii. The closed statements follow by De Morgan's laws. (Infinite intersections of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.) ∎

Proposition 2.3

(a) $x \in A'$ iff there is a sequence in $A \setminus \{x\}$ converging to x; and then every ball about x contains infinitely many points of A. (b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) Pick $a_k \in N(x,1/k)\cap A$, $a_k \neq x$. If a ball $N(x,r)$ met $A\setminus\{x\}$ in finitely many points, a smaller ball would miss them all. (b) Follows from (a), using constant sequences for points of A. ∎

Theorem 2.4

For $F \subset \mathbb{R}^n$ the following are equivalent: (a) F is closed; (b) $F' \subset F$; (c) whenever $x_k \in F$ and $x_k \to x$, then $x \in F$.

Proof. (a)⇒(b): if $x \notin F$, some ball about x misses F, so $x \notin F'$. (b)⇒(c): a limit of points of F is in F or is a limit point of F (Proposition 2.3). (c)⇒(a): if $x \notin F$ and every ball $N(x,1/k)$ met F, we would get a sequence in F converging to $x \notin F$. ∎

In particular $\overline{A}$ is the smallest closed set containing A, and A' is always closed: if x is a limit point of A', each ball about x contains a point of A', hence (by shrinking) infinitely many points of A.

A set $Y \subset X$ is **open in** X (relatively open) if $Y = X \cap U$ for some open $U \subset \mathbb{R}^n$; equivalently, every $y \in Y$ has a ball with $N(y,r)\cap X \subset Y$. Relatively closed sets are defined likewise. This is the right notion when functions are defined only on X.

2.3 Bolzano–Weierstrass and Cauchy sequences

A sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges, because of the comparison between $\|\cdot\|$ and coordinates.

Theorem 2.5 (Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point. Equivalently, every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. Enclose A in a closed box B_0 (a product of closed intervals). Bisect each side to split B_0 into 2^n boxes; one of them, B_1, contains infinitely many points of A. Repeat to get boxes $B_0 \supset B_1 \supset \cdots$, each containing infinitely many points of A, with diameters $\to 0$. Applying the nested interval theorem (Theorem 1.11) in each coordinate gives a point $x \in \bigcap B_k$, and every ball about x contains some B_k, hence infinitely many points of A. For a bounded sequence with infinitely many distinct values, choose terms closer and closer to such a limit point with increasing indices; if it has finitely many values, some value repeats infinitely often. ∎

For a bounded real sequence, let A be the set of limits of its convergent subsequences (subsequential limits). Then $\limsup x_n = \max A$ and $\liminf x_n = \min A$: by Proposition 1.9, $x_n > \limsup - \epsilon$ infinitely often yields a subsequence converging to lim sup, and no subsequential limit can exceed it.

Definition (Cauchy sequence)

(x_k) in $\mathbb{R}^n$ is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_k - x_l\| < \epsilon$ for all $k,l \geq N$.

Theorem 2.6 (Cauchy criterion)

A sequence in $\mathbb{R}^n$ converges iff it is Cauchy.

Proof. Convergent sequences are Cauchy ($\|x_k - x_l\| \leq \|x_k - x\| + \|x - x_l\|$). A Cauchy sequence is bounded (take $\epsilon = 1$), so by Bolzano–Weierstrass a subsequence $x_{k_j} \to x$. Given ϵ, choose N from the Cauchy condition and j with $k_j \geq N$ and $\|x_{k_j} - x\| < \epsilon$; then $\|x_k - x\| < 2\epsilon$ for $k \geq N$. ∎

Applied to partial sums: $\sum a_k$ converges iff for every $\epsilon > 0$ there is N with $\|\sum_{k=m}^{n} a_k\| < \epsilon$ for $n \geq m \geq N$. In particular absolute convergence ($\sum\|a_k\| < \infty$) implies convergence, and $a_k \to 0$ is

necessary for convergence.

2.4 Convergence tests for series

Proposition 2.7 (Rearrangements)

If $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ is a bijection, then $\sum a_{\tau(n)} = \sum a_n$ (both finite or both infinite). The same holds for absolutely convergent series of reals.

Proof. Every partial sum of $\sum a_{\tau(n)}$ is at most $\sum a_n$, and conversely, so the suprema agree. For an absolutely convergent series write $a_n = a_n^+ - a_n^-$ with $a_n^{\pm} \geq 0$ and apply the first part to each. ∎

Conditionally convergent series behave differently: by Riemann's theorem a convergent but not absolutely convergent real series can be rearranged to converge to any prescribed value (take positive terms until the target is exceeded, then negative terms, and so on; the terms tend to 0).

Theorem 2.8 (Alternating series)

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and the error after n terms is at most x_{n+1}.

Proof. The even partial sums s_{2m} increase, the odd ones s_{2m+1} decrease, $s_{2m} \leq s_{2m+1}$, and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$. Nested intervals $[s_{2m}, s_{2m+1}]$ shrink to the sum. ∎

Theorem 2.9 (Comparison, root and ratio tests)

(a) If $0 \leq a_n \leq b_n$ and $\sum b_n < \infty$, then $\sum a_n < \infty$. (b) Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, it diverges. (c) If $\limsup |a_{n+1}/a_n| < 1$ the series converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large n, it diverges.

Proof. (a) Bounded partial sums. (b) If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n; compare with the geometric series. If $\alpha > 1$, $|a_n| > 1$ infinitely often, so $a_n \not\to 0$. (c) If the ratios are eventually $< \beta < 1$, then $|a_n| \leq C\beta^n$. ∎

For positive c_n,

$$\liminf \frac{c_{n+1}}{c_n} \leq \liminf c_n^{1/n} \leq \limsup c_n^{1/n} \leq \limsup \frac{c_{n+1}}{c_n},$$

so the root test is stronger than the ratio test (whenever the ratio test decides, so does the root test). When $\alpha = 1$ neither test decides: $\sum 1/n$ diverges and $\sum 1/n^2$ converges.

2.5 Compact sets

Definition (Compactness)

An **open cover** of K is a family of open sets whose union contains K. K is **compact** if every open cover has a finite subcover.

Proposition 2.10

A compact set is closed and bounded. A closed subset of a compact set is compact.

Proof. The balls $N(0,k)$ cover K; a finite subcover gives boundedness. If $x \notin K$, the sets $U_k = \{y : \|y-x\| > 1/k\}$ cover K, so finitely many do, and then a ball $N(x,1/k)$ misses K: K^c is open. For $F \subset K$ closed, add F^c to any open cover of F to get a cover of K. ■

Theorem 2.11

Every closed box $B = [a_1,b_1] \times \cdots \times [a_n,b_n]$ is compact.

Proof. Suppose an open cover has no finite subcover of B. Bisecting as in Theorem 2.5, one of the 2^n sub-boxes has no finite subcover; repeat to get nested boxes B_k with diameters $\to 0$, none finitely covered. They share a point x, which lies in some open set U of the cover; but $U \supset N(x,r) \supset B_k$ for large k, a contradiction. ■

Theorem 2.12 (Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent: (a) K is compact; (b) K is closed and bounded; (c) every sequence in K has a subsequence converging to a point of K; (d) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b) is Proposition 2.10; (b)⇒(a) since K is a closed subset of a compact box. (b)⇒(c): Bolzano–Weierstrass plus closedness (Theorem 2.4). (c)⇒(d): pick distinct points of the infinite set. (d)⇒(b): if K were unbounded, points x_k with $\|x_k\| > k$ would form an infinite set with no limit point; if some $x \in K' \setminus K$, points of K converging to x would have x as their only limit point, which is not in K. ■

Theorem 2.13 (Finite intersection property)

If $\{K_i\}$ is a family of compact sets such that every finite subfamily has nonempty intersection, then $\bigcap_i K_i \neq \emptyset$.

Proof. Fix K_1. If $\bigcap K_i = \emptyset$, the open sets K_i^c cover K_1, so finitely many do: $K_1 \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

Example 2.1

(a) A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point. Otherwise $S = \{s_1, s_2, \dots\}$ is perfect; choose closed balls $\overline{N}_1 \supset \overline{N}_2 \supset \cdots$ with centers in S, radii $\to 0$, each meeting S, and $s_k \notin \overline{N}_k$ (possible because no point is isolated). The compact sets $\overline{N}_k \cap S$ are nested and nonempty, so they share a point of S, which differs from every s_k. Hence, e.g., the Cantor set (closed, no isolated points) is uncountable.
(b) $(0,1]$ is not compact: the cover $\{(1/k, 2) : k \in \mathbb{N}\}$ has no finite subcover.

2.6 Connected sets

Definition (Connectedness)

$S \subset \mathbb{R}^n$ is **disconnected** if there are open sets U, V with $S \subset U \cup V$, $S \cap U \neq \emptyset \neq S \cap V$ and $S \cap U \cap V = \emptyset$; otherwise S is **connected**.

Proposition 2.14

A subset of $\mathbb{R}$ is connected iff it is an interval (i.e. $x < z < y$ with $x, y \in S$ implies $z \in S$).

Proof. If S is not an interval, some $z \notin S$ lies between points of S, and $(-\infty, z)$, (z, ∞) disconnect S. Conversely let S be an interval separated by U, V, with $a \in S \cap U$, $b \in S \cap V$, say $a < b$. Let $c = \sup\{x \in [a, b] : x \in U\}$; $c \in S$. If $c \in U$, then $c < b$ and, U being open, points slightly larger than c in $[a, b]$ lie in U, contradicting the definition of c. If $c \in V$, then $c > a$ and a neighbourhood of c lies in V, so points slightly smaller than c are not in U and some smaller number is an upper bound: contradiction. ■

Proposition 2.15

If connected sets C_i have a common point, then $\bigcup C_i$ is connected. A path-connected set (any two points joined by a continuous path in the set) is connected, and an open connected subset of $\mathbb{R}^n$ is path-connected.

Proof. If U, V separated the union, each C_i would lie entirely in U or entirely in V (by connectedness of C_i), and the common point forces all into the same one. Path-connectedness implies connectedness by Theorem 3.7 in Chapter 3 (continuous images of intervals are connected) and the first statement applied to the paths from a fixed point. For an open connected G, the set of points reachable by polygonal paths from a fixed x_0 is open, and so is its complement in G (balls are convex); connectedness forces it to be all of G. ■

Convex sets, in particular balls and boxes, are connected. The closure of a connected set is connected.

Chapter 3. Continuous Functions

Throughout, $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$, and $f : X \to Y$.

3.1 Limits and continuity

Definition (Limit and continuity)

Let $x_0 \in X'$. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ such that $x \in X$, $0 < \|x - x_0\| < \delta$ imply $\|f(x) - y_0\| < \epsilon$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$. It is continuous on X if continuous at every point.

At an isolated point every function is continuous; at a limit point $x_0 \in X$, continuity means $\lim_{x\to x_0} f(x) = f(x_0)$.

Proposition 3.1 (Sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X. Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for all $x_k \to x_0$ with $x_k \neq x_0$.

Proof. Necessity is immediate from the definitions. If f is not continuous at x_0, there is $\epsilon > 0$ such that for each k some x_k with $\|x_k - x_0\| < 1/k$ has $\|f(x_k) - f(x_0)\| \geq \epsilon$; this sequence violates the criterion. ∎

Consequently sums, scalar multiples, products and quotients (where defined, nonzero denominator) of continuous functions are continuous, and compositions of continuous functions are continuous. A map into $\mathbb{R}^m$ is continuous iff each component is. Polynomials in n variables are continuous.

Theorem 3.2 (Topological characterization)

The following are equivalent: (a) $f : X \to Y$ is continuous; (b) $f^{-1}(V)$ is open in X for every open $V \subset \mathbb{R}^m$; (c) $f^{-1}(F)$ is closed in X for every closed $F \subset \mathbb{R}^m$.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$, choose ϵ with $N(f(x), \epsilon) \subset V$ and the corresponding δ; then $N(x, \delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): apply (b) to $V = N(f(x_0), \epsilon)$. (b)⇔(c) by taking complements. ∎

Example 3.1

(a) $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$, is continuous in each variable separately but not at the origin: along $y = mx$ it equals $m/(1 + m^2)$.
(b) The function equal to $1/q$ at $x = p/q$ in lowest terms ($q \geq 1$) and 0 at irrationals is continuous exactly at the irrationals: near any x, only finitely many rationals have denominator $\leq q$.
(c) $\{x : f(x) > 0\}$ is open for continuous real f on $\mathbb{R}^n$; $\{x : f(x) = 0\}$ is closed.

3.2 Compactness, connectedness and continuity

Theorem 3.3

The continuous image of a compact set is compact.

Proof. If $\{V_i\}$ is an open cover of $f(X)$, the sets $f^{-1}(V_i)$ are relatively open and cover X: $f^{-1}(V_i) = X \cap U_i$ with U_i open. Finitely many U_i cover X, so finitely many V_i cover $f(X)$. (Alternatively, use sequential compactness.) ■

Theorem 3.4 (Extreme value theorem)

A continuous real function on a nonempty compact set attains a maximum and a minimum. In particular, if $f > 0$ on compact X, then $f \geq c$ for some constant $c > 0$.

Proof. $f(X)$ is compact, hence closed and bounded, so it contains $\sup f(X)$ and $\inf f(X)$. ■

Proposition 3.5

If $f : X \to Y$ is a continuous bijection and X is compact, then f^{-1} is continuous.

Proof. For $F \subset X$ closed, F is compact, so $(f^{-1})^{-1}(F) = f(F)$ is compact, hence closed. Apply Theorem 3.2(c). ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection from $[0, 2\pi)$ onto the circle whose inverse is discontinuous at $(1, 0)$.

Proposition 3.6

If $A \subset \mathbb{R}^n$ is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\} > 0$.

Proof. $x \mapsto \operatorname{dist}(x, B)$ is continuous (indeed $|\operatorname{dist}(x, B) - \operatorname{dist}(y, B)| \leq \|x - y\|$) and positive on A since B is closed; apply Theorem 3.4. (Both sets merely closed is not enough: consider the x-axis and the graph of $1/x$.) ■

Theorem 3.7

The continuous image of a connected set is connected.

Proof. If open U, V separated $f(X)$, then $f^{-1}(U), f^{-1}(V)$ would separate X. ■

Corollary 3.8 (Intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \gamma < f(b)$, then $f(c) = \gamma$ for some $c \in (a, b)$. The continuous image of an interval is an interval, and the image of $[a, b]$ is a closed bounded interval.

Proof. $f([a, b])$ is connected, hence an interval (Proposition 2.14); combined with Theorem 3.3 it is $[\min f, \max f]$. ■

Example 3.2

(a) Every continuous $f : [a, b] \to [a, b]$ has a fixed point: apply the intermediate value theorem to $f(x) - x$.

(b) Every polynomial of odd degree has a real root. Every $y > 0$ has a unique positive n-th root.
(c) No continuous $f : \mathbb{R} \to \mathbb{R}$ takes every value exactly twice. (Consider the maximum on the interval between the two preimages of some value.)

3.3 Uniform continuity

Definition (Uniform continuity)

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ such that $x, x' \in X$, $\|x - x'\| < \delta$ imply $\|f(x) - f(x')\| < \epsilon$.

The difference from continuity is that δ may not depend on the point. $x \mapsto 1/x$ on $(0, 1)$ and $x \mapsto x^2$ on $\mathbb{R}$ are continuous but not uniformly continuous; Lipschitz functions ($\|f(x) - f(x')\| \le L\|x - x'\|$) are uniformly continuous.

Theorem 3.9 (Heine)

A continuous function on a compact set is uniformly continuous.

Proof. If not, there are $\epsilon > 0$ and x_k, x'_k with $\|x_k - x'_k\| < 1/k$ but $\|f(x_k) - f(x'_k)\| \ge \epsilon$. By compactness a subsequence $x_{k_j} \to x \in X$, and then $x'_{k_j} \to x$ too; continuity at x gives $f(x_{k_j}) - f(x'_{k_j}) \to 0$, a contradiction. ■

Continuous maps preserve convergent sequences but not Cauchy sequences ($1/x$ maps the Cauchy sequence $1/k$ in $(0, 1)$ to an unbounded one). Uniformly continuous maps do preserve them:

Theorem 3.10

If $f : X \to \mathbb{R}^m$ is uniformly continuous, it maps Cauchy sequences to Cauchy sequences and extends uniquely to a continuous (indeed uniformly continuous) function on $\overline{X}$.

Proof. The first claim is immediate. For $x \in \overline{X}$ choose $x_k \to x$ in X; $(f(x_k))$ is Cauchy, hence convergent, and the limit does not depend on the sequence (interleave two sequences). Define $\tilde{f}(x)$ as this limit; the ϵ–δ estimate passes to the limit. ■

So a continuous function on a bounded set X is uniformly continuous iff it extends continuously to the compact set $\overline{X}$.

3.4 Monotone functions

Theorem 3.11

A monotone function f on an open interval (a, b) has one-sided limits $f(x-)$ and $f(x+)$ at every point. If f is increasing, $f(x-) \le f(x) \le f(x+)$, and $f(x+) \le f(y-)$ for $x < y$.

Proof. For increasing f, $f(x-) = \sup\{f(t) : t < x\}$: given ϵ, some $t_0 < x$ has $f(t_0) > f(x-) - \epsilon$, and monotonicity gives the estimate for all $t \in (t_0, x)$. Similarly $f(x+) = \inf_{t>x} f(t)$. ■

Corollary 3.12

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, at each discontinuity x choose a rational $r(x) \in (f(x-), f(x+))$. These open intervals are disjoint for different x (by the last inequality of Theorem 3.11), so $x \mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{q_k\}$ the increasing function $\sum_k 2^{-k}\mathbf{1}\{x > q_k\}$ is discontinuous exactly at the q_k.

Theorem 3.13

For a continuous f on an interval I the following are equivalent: (a) f is injective; (b) f is strictly monotone. In that case $f(I)$ is an interval and $f^{-1} : f(I) \to I$ is continuous and strictly monotone.

Proof. (a)⇒(b): if f were not monotone, there would be $x < y < z$ with $f(y)$ not between $f(x)$ and $f(z)$, and the intermediate value theorem would produce a repeated value. For the inverse: f^{-1} is strictly monotone on the interval $f(I)$, and its image I is an interval, so it has no jumps (a jump would leave a gap in the image), i.e. it is continuous. ■

This gives continuity of $x^{1/n}$, $\log x$, $\arcsin x$, etc., from that of their inverses.

Chapter 4. Differentiable Functions

4.1 Differentiability

Definition (Derivative)

$f:(a,b)\to\mathbb{R}$ is **differentiable at** x_0 if $f'(x_0)=\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}$ exists.

Proposition 4.1 (Linear approximation)

f is differentiable at x_0 iff there are a number a and a function η with $\eta(h)\to 0=\eta(0)$ such that

$$f(x_0+h)=f(x_0)+ah+|h|\eta(h) \qquad \text{for small } h.$$

Then $a=f'(x_0)$. Equivalently, there is ϕ continuous at x_0 with $f(x)-f(x_0)=\phi(x)(x-x_0)$, and $\phi(x_0)=f'(x_0)$.

This form (the one that generalizes to several variables) gives the rules at once. A differentiable function is continuous. Sums, products and quotients obey the usual rules.

Proposition 4.2 (Chain rule)

If f is differentiable at x_0 and g at $y_0=f(x_0)$, then $g\circ f$ is differentiable at x_0 with $(g\circ f)'(x_0)=g'(y_0)f'(x_0)$.

Proof. Write $f(x)-f(x_0)=\phi(x)(x-x_0)$ and $g(y)-g(y_0)=\psi(y)(y-y_0)$ with ϕ,ψ continuous at x_0,y_0. Then $g(f(x))-g(f(x_0))=\psi(f(x))\phi(x)(x-x_0)$, and $\psi(f(x))\phi(x)\to g'(y_0)f'(x_0)$. (The naive proof dividing by $f(x)-f(x_0)$ fails when this vanishes.) ■

Example 4.1

(a) $f(x)=x^2\sin(1/x)$, $f(0)=0$, is differentiable everywhere with $f'(0)=0$, but $f'(x)=2x\sin(1/x)-\cos(1/x)$ is discontinuous at 0.
(b) If f is differentiable at x_0 with $f'(x_0)\neq 0$ and invertible near x_0 with continuous inverse, then $(f^{-1})'(f(x_0))=1/f'(x_0)$.

4.2 Mean value theorems

If f has a local maximum or minimum at an interior point c where it is differentiable, then $f'(c)=0$ (the difference quotients have opposite signs on the two sides).

Theorem 4.3 (Cauchy's mean value theorem)

If f,g are continuous on $[a,b]$ and differentiable on (a,b), there is $c\in(a,b)$ with

$$\{f(b)-f(a)\}g'(c)=\{g(b)-g(a)\}f'(c).$$

Proof. $h(x)=\{f(b)-f(a)\}g(x)-\{g(b)-g(a)\}f(x)$ satisfies $h(a)=h(b)$. If h is constant any c works; otherwise it attains a maximum or minimum at an interior point c, where $h'(c)=0$ (Rolle's theorem). ■

With $g(x) = x$ this is the **mean value theorem**: $f(b) - f(a) = f'(c)(b-a)$. Hence a differentiable f on (a,b) is increasing, constant or decreasing according as $f' \geq 0$, $f' = 0$ or $f' \leq 0$; and $|f'| \leq L$ makes f Lipschitz with constant L.

Theorem 4.4 (L'Hôpital's rule)

Let f, g be differentiable on (a,b), $g' \neq 0$, and $\lim_{x\to a+} f(x) = \lim_{x\to a+} g(x) = 0$ (or $|g(x)| \to \infty$). If $f'(x)/g'(x) \to L$ as $x \to a+$, then $f(x)/g(x) \to L$.

Proof. In the 0/0 case extend $f(a) = g(a) = 0$; for x near a, Cauchy's theorem gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with $a < c_x < x$. In the ∞ case fix y close to a with $|f'/g' - L| < \epsilon$ on (a,y) and write, for $a < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$; then $\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}(1 - \frac{g(y)}{g(x)}) + \frac{f(y)}{g(x)}$, and the last two corrections vanish as $x \to a$. ■

Theorem 4.5 (Darboux)

If F is differentiable on $[a,b]$ and $F'(a) < \gamma < F'(b)$, then $F'(c) = \gamma$ for some $c \in (a,b)$.

Proof. $G(x) = F(x) - \gamma x$ has $G'(a) < 0 < G'(b)$, so G takes values below $G(a)$ just right of a and below $G(b)$ just left of b. Its minimum on $[a,b]$ is therefore attained at an interior point c, where $G'(c) = 0$. ■

So derivatives have the intermediate value property although they need not be continuous: a function with a jump discontinuity (such as $\operatorname{sgn} x$) has no antiderivative on an interval.

4.3 Taylor's theorem

Theorem 4.6 (Taylor)

Let I be the closed interval with endpoints $a \neq b$, and suppose $f^{(n)}$ is continuous on I and $f^{(n+1)}$ exists in its interior. Then

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(b-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}$$

for some c strictly between a and b (Lagrange form of the remainder).

Proof. Let $P(x) = \sum_{k\leq n} f^{(k)}(x)(b-x)^k/k!$, so $P(b) = f(b)$ and, telescoping, $P'(x) = f^{(n+1)}(x)(b-x)^n/n!$. Apply Cauchy's mean value theorem to P and $g(x) = (b-x)^{n+1}$ on I: $\{P(b)-P(a)\}g'(c) = \{g(b) - g(a)\}P'(c)$, i.e. $\{f(b) - P(a)\}(n+1)(b-c)^n = (b-a)^{n+1}f^{(n+1)}(c)(b-c)^n/n!$. ■

With $g(x) = b - x$ instead one gets the Cauchy form $\frac{f^{(n+1)}(c)}{n!}(b-c)^n(b-a)$; and if $f^{(n+1)}$ is Riemann integrable, the integral form $\frac{1}{n!}\int_a^b f^{(n+1)}(t)(b-t)^n dt$ follows from Chapter 5.

Proposition 4.7 (When the Taylor series converges)

Let f be C^∞ on an interval $J \ni a$. If there is M with $|f^{(k)}(x)| \leq M$ for all $x \in J$ and all k, then the Taylor series of f at a converges to f on J.

Proof. The Lagrange remainder is at most $M|x-a|^{n+1}/(n+1)! \to 0$. ■

Example 4.2

(a) $e^x = \sum x^k/k!$, $\sin x = \sum(-1)^k x^{2k+1}/(2k+1)!$ for all x, since all derivatives are bounded on bounded intervals.
(b) $f(x) = e^{-1/x^2}$ $(x \neq 0)$, $f(0) = 0$, is C^∞ with $f^{(k)}(0) = 0$ for all k (each derivative is a polynomial in $1/x$ times e^{-1/x^2}). Its Taylor series at 0 is identically 0 and differs from f at every $x \neq 0$: C^∞ does not imply analytic.
(c) Second derivative test: if $f'(c) = 0$ and f'' is continuous with $f''(c) > 0$, then c is a strict local minimum, by Taylor's theorem with $n = 1$.

Chapter 5. The Riemann–Stieltjes Integral

5.1 The Riemann integral

Let $f:[a,b]\to\mathbb{R}$ be bounded. A **partition** $P=\{a=x_0<x_1<\cdots<x_n=b\}$ has mesh $\|P\|=\max\Delta x_i$, $\Delta x_i=x_i-x_{i-1}$. With $M_i=\sup_{[x_{i-1},x_i]}f$ and $m_i=\inf_{[x_{i-1},x_i]}f$, the upper and lower sums are

$$U(f,P)=\sum M_i\Delta x_i,\qquad L(f,P)=\sum m_i\Delta x_i.$$

Proposition 5.1 (Refinement)

If $P\subset Q$, then $L(f,P)\le L(f,Q)\le U(f,Q)\le U(f,P)$. Hence $L(f,P)\le U(f,P')$ for any two partitions.

Proof. Adding one point splits one interval; the infimum over each half is at least the infimum over the whole, so L cannot decrease; similarly for U. For the second claim compare both with $P\cup P'$. ∎

Definition (Riemann integrability)

The lower and upper integrals are $\underline{\int}f=\sup_P L(f,P)$ and $\overline{\int}f=\inf_P U(f,P)$; always $\underline{\int}f\le\overline{\int}f$. f is **Riemann integrable**, $f\in\mathcal{R}[a,b]$, if they are equal; the common value is $\int_a^b f$.

Proposition 5.2 (Riemann's criterion)

$f\in\mathcal{R}[a,b]$ iff for every $\epsilon>0$ there is a partition P with $U(f,P)-L(f,P)<\epsilon$.

Proof. If f is integrable, choose P_1,P_2 with $U(f,P_1)<\int f+\epsilon/2$, $L(f,P_2)>\int f-\epsilon/2$, and take $P=P_1\cup P_2$. Conversely $0\le\overline{\int}f-\underline{\int}f\le U(f,P)-L(f,P)<\epsilon$. ∎

The Dirichlet function (1 on $\mathbb{Q}$, 0 otherwise) is not integrable: every $U=1$ and every $L=0$.

Theorem 5.3 (Properties)

For $f,g\in\mathcal{R}[a,b]$ and $c\in\mathbb{R}$: $f+g$, $cf\in\mathcal{R}$ and the integral is linear; $f\le g$ implies $\int f\le\int g$; for $a<c<b$, f is integrable on $[a,c]$ and $[c,b]$ with $\int_a^b=\int_a^c+\int_c^b$; $|f|\in\mathcal{R}$ and $|\int f|\le\int|f|$; $fg\in\mathcal{R}$.

Proof. For sums, $\sup(f+g)\le\sup f+\sup g$ on each subinterval, so $U(f+g,P)\le U(f,P)+U(g,P)$, similarly for L, and Riemann's criterion applies. For $|f|$, $\sup|f|-\inf|f|\le\sup f-\inf f$ on each subinterval. For products, f^2 is integrable because $\sup f^2-\inf f^2\le 2K(\sup f-\inf f)$ with $K=\sup|f|$, and $fg=\frac14\{(f+g)^2-(f-g)^2\}$. Additivity: insert c into partitions. ∎

5.2 Integrable functions

Theorem 5.4

Continuous functions and monotone functions on $[a,b]$ are Riemann integrable.

Proof. Continuous: by uniform continuity (Theorem 3.9) choose δ with oscillation $<\epsilon/(b-a)$ on intervals shorter than δ; any P with $\|P\|<\delta$ gives $U-L<\epsilon$. Increasing: with the uniform partition into n pieces, $U-L=\frac{b-a}{n}\sum(f(x_i)-f(x_{i-1}))=\frac{b-a}{n}(f(b)-f(a))\to 0$. ∎

More generally a bounded function with finitely many discontinuities is integrable (cover the discontinuities by short intervals), and if $f \in \mathcal{R}$ and ϕ is continuous on the range of f, then $\phi \circ f \in \mathcal{R}$. (The complete answer — f is integrable iff its set of discontinuities has measure zero — is Lebesgue's criterion, Chapter 10.)

Riemann's original definition uses **Riemann sums** $S(f,P,\xi) = \sum f(\xi_i)\Delta x_i$ with tags $\xi_i \in [x_{i-1}, x_i]$.

Theorem 5.5

For bounded f and $A \in \mathbb{R}$ the following are equivalent: (a) $f \in \mathcal{R}[a,b]$ and $\int f = A$; (b) for every $\epsilon > 0$ there is $\delta > 0$ such that $|S(f,P,\xi) - A| < \epsilon$ for all tagged partitions with $\|P\| < \delta$.

Proof. (b)⇒(a): choosing tags near the sup or inf on each piece shows $U(f,P)$ and $L(f,P)$ are within 2ϵ of A. (a)⇒(b): fix P_0 with $U(f,P_0) - L(f,P_0) < \epsilon$, having k points, and $|f| \le K$. If $\|P\| < \delta = \epsilon/(kK)$, at most k intervals of P contain points of P_0, contributing at most $2kK\delta = 2\epsilon$ to $U(f,P) - L(f,P)$; the remaining intervals lie inside intervals of P_0 and contribute at most $U(f,P_0) - L(f,P_0)$. So $U(f,P) - L(f,P) < 3\epsilon$, and every Riemann sum lies between $L(f,P)$ and $U(f,P)$. ∎

Example 5.1

$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{n^2+k^2} = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$, recognizing a Riemann sum with mesh $1/n$.

5.3 The fundamental theorem of calculus

Theorem 5.6

For $f \in \mathcal{R}[a,b]$, $F(x) = \int_a^x f$ is Lipschitz, hence uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 with $F'(x_0) = f(x_0)$.

Proof. $|F(y)-F(x)| \le K|y-x|$ with $K = \sup|f|$. Next, $\frac{F(x_0+h)-F(x_0)}{h} - f(x_0) = \frac{1}{h}\int_{x_0}^{x_0+h}(f - f(x_0))$, which is at most $\sup_{|t-x_0|\le|h|}|f(t) - f(x_0)| \to 0$ in absolute value. ∎

So every continuous function has an antiderivative.

Theorem 5.7 (Fundamental theorem of calculus)

If F is differentiable on $[a,b]$ and $f = F' \in \mathcal{R}[a,b]$, then $\int_a^b f = F(b) - F(a)$.

Proof. For any partition, the mean value theorem gives $F(b) - F(a) = \sum F'(\xi_i)\Delta x_i$ for suitable ξ_i, a Riemann sum lying between $L(f,P)$ and $U(f,P)$. Hence $\underline{\int} f \le F(b) - F(a) \le \overline{\int} f$. ∎

The hypothesis $F' \in \mathcal{R}$ cannot be dropped: there are differentiable F with bounded but non-integrable derivatives (Volterra), and $x^2 \sin(1/x^2)$ has an unbounded derivative.

Proposition 5.8 (Substitution and integration by parts)

(a) If $g : [a,b] \to [c,d]$ is C^1 and f is continuous on $[c,d]$, then $\int_a^b f(g(t))g'(t)dt = \int_{g(a)}^{g(b)} f(x)dx$.
(b) If F, G are differentiable with $F' = f$, $G' = g$ integrable, then $\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG$.

Proof. (a) With $\Phi(y) = \int_{g(a)}^{y} f$, $(\Phi \circ g)' = f(g)g'$; apply Theorem 5.7. (b) Apply Theorem 5.7 to FG, whose derivative $fG + Fg$ is integrable. ∎

5.4 Functions of bounded variation

Definition (Bounded variation)

The **total variation** of $f : [a, b] \to \mathbb{R}$ is $V_a^b(f) = \sup_P \sum_i |f(x_i) - f(x_{i-1})|$. f is of **bounded variation** (BV) if $V_a^b(f) < \infty$.

Monotone functions are BV with $V = |f(b) - f(a)|$; Lipschitz functions are BV; $x \sin(1/x)$ is continuous on $[0, 1]$ but not BV (the variation over the points $1/((k + \frac{1}{2})\pi)$ is a harmonic series). Sums and products of BV functions are BV, and $V_a^b = V_a^c + V_c^b$ for $a < c < b$ (refine partitions by c).

Theorem 5.9 (Jordan decomposition)

f is BV on $[a, b]$ iff $f = g - h$ with g, h increasing. One can take $g(x) = V_a^x(f)$ and $h = g - f$.

Proof. Differences of increasing functions are BV. Conversely $g(x) = V_a^x(f)$ increases, and for $x < y$, $h(y) - h(x) = V_x^y(f) - (f(y) - f(x)) \geq 0$. ∎

Hence a BV function has one-sided limits everywhere and countably many discontinuities, and is Riemann integrable.

A curve $\gamma : [a, b] \to \mathbb{R}^m$ has length $\ell(\gamma) = \sup_P \sum \|\gamma(t_i) - \gamma(t_{i-1})\|$; it is **rectifiable** if this is finite. Since $|\gamma_j(t) - \gamma_j(s)| \leq \|\gamma(t) - \gamma(s)\| \leq \sum_j |\gamma_j(t) - \gamma_j(s)|$, γ is rectifiable iff every component is BV.

Theorem 5.10

A C^1 curve $\gamma : [a, b] \to \mathbb{R}^m$ is rectifiable with $\ell(\gamma) = \int_a^b \|\gamma'(t)\| dt$.

Proof. $\|\gamma(t_i) - \gamma(t_{i-1})\| = \|\int_{t_{i-1}}^{t_i} \gamma'\| \leq \int_{t_{i-1}}^{t_i} \|\gamma'\|$ gives $\ell \leq \int \|\gamma'\|$. Conversely, by uniform continuity of γ', for fine partitions γ' varies by less than ϵ on each piece, so $\|\gamma(t_i) - \gamma(t_{i-1})\| \geq (\|\gamma'(t_i)\| - \epsilon)\Delta t_i$, and summing gives $\ell \geq \int \|\gamma'\| - 2\epsilon(b - a)$ in the limit. ∎

5.5 The Riemann–Stieltjes integral

Let $\alpha : [a, b] \to \mathbb{R}$ be increasing. Replace Δx_i by $\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1})$: $U(f, \alpha, P) = \sum M_i \Delta \alpha_i$, $L(f, \alpha, P) = \sum m_i \Delta \alpha_i$. f is **Riemann–Stieltjes integrable** with respect to α, $f \in \mathcal{R}(\alpha)$, if $\inf_P U = \sup_P L$; the value is $\int_a^b f \, d\alpha$. Refinement, Riemann's criterion and linearity in f and in α carry over word for word; for BV $\alpha = \alpha_1 - \alpha_2$ (increasing parts) one sets $\int f d\alpha = \int f d\alpha_1 - \int f d\alpha_2$, which is equivalent to the Riemann-sum definition with mesh $\to 0$ when f is continuous.

Proposition 5.11

If f is continuous on $[a, b]$ and α is BV, then $f \in \mathcal{R}(\alpha)$, and $|\int f d\alpha| \leq \sup|f| \cdot V_a^b(\alpha)$.

Proof. For increasing α: choose δ from uniform continuity so that the oscillation of f on pieces shorter than δ is less than ϵ; then $U - L \leq \epsilon(\alpha(b) - \alpha(a))$. ∎

Theorem 5.12

(a) If α is C^1 and f is bounded, then $f \in \mathcal{R}(\alpha)$ iff $f\alpha' \in \mathcal{R}$, and $\int f d\alpha = \int f(x)\alpha'(x)dx$. (b) If α is a step function with jumps c_k at points s_k in the interior and f is continuous at each s_k, then $\int f d\alpha = \sum_k c_k f(s_k)$.

Proof. (a) By the mean value theorem $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$, so Riemann–Stieltjes sums of f and Riemann sums of $f\alpha'$ with the same tags differ by $\sum f(\xi_i)\{\alpha'(t_i) - \alpha'(\xi_i)\}\Delta x_i$, which is small by uniform continuity of α'. (b) Only the subintervals containing a jump contribute, and there $f \approx f(s_k)$. ∎

Thus sums and integrals are special cases of one object; in probability, $E\,g(X) = \int g\,dF$ for a distribution function F.

Theorem 5.13 (Integration by parts)

If f, α are BV and $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Summation by parts: for a tagged partition, $\sum \alpha(\xi_i)\{f(x_i) - f(x_{i-1})\} = f(b)\alpha(b) - f(a)\alpha(a) - \sum f(x_i)\{\alpha(\xi_{i+1}) - \alpha(\xi_i)\}$ (with $\xi_0 = a$, $\xi_{n+1} = b$), and the last sum is a Riemann–Stieltjes sum for $\int f d\alpha$ over the partition $\{\xi_i\}$, whose mesh is at most $2\|P\|$. ∎

Corollary 5.14 (Second mean value theorem)

If f is increasing and g is continuous on $[a, b]$, there is $c \in [a, b]$ with $\int_a^b f(x)g(x)dx = f(a)\int_a^c g + f(b)\int_c^b g$.

Proof. With $G(x) = \int_a^x g$, Theorem 5.12(a) and integration by parts give $\int fg = \int f dG = f(b)G(b) - \int_a^b G\,df$. Since $df \geq 0$ and G is continuous, $\int G df = G(c)(f(b) - f(a))$ for some c (intermediate value theorem). Rearranging gives the formula. ∎

Appendix A. The Number Systems

Chapter 1 took the real numbers as a complete ordered field. Here we sketch how such a field is built from the natural numbers, so that the axioms are consistent.

A.1 Natural numbers and induction

The natural numbers are characterized by Peano's axioms: there is a set $\mathbb{N}$ with an element 1 and an injective successor map $s : \mathbb{N} \to \mathbb{N}$ with $1 \notin s(\mathbb{N})$, such that any subset containing 1 and closed under s is all of $\mathbb{N}$ (**induction**). Addition and multiplication are defined recursively ($m+1 = s(m)$, $m + s(n) = s(m + n)$; $m \cdot 1 = m$, $m \cdot s(n) = mn + m$), and their associative, commutative and distributive laws, as well as the order $m < n \iff n = m + k$ for some k, are proved by induction. Induction is equivalent to the **well-ordering principle**: every nonempty subset of $\mathbb{N}$ has a least element.

A.2 Integers and rationals

$\mathbb{Z}$ is obtained from pairs $(m, n) \in \mathbb{N}\times\mathbb{N}$ (thought of as $m-n$) modulo $(m, n) \sim (m', n') \iff m+n' = m'+n$, with $(m, n)+(m', n') = (m+m', n+n')$ and $(m, n)(m', n') = (mm'+nn', mn'+nm')$; these operations are well defined on classes, and $\mathbb{Z}$ is a commutative ring with an order. $\mathbb{Q}$ is obtained from pairs $(p, q) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ (thought of as p/q) modulo $(p, q) \sim (p', q') \iff pq' = p'q$; it is an ordered field (positive elements: classes with $pq > 0$) which is Archimedean but not complete.

A.3 Real numbers by Dedekind cuts

Definition (Cut)

A **cut** is a subset $\alpha \subset \mathbb{Q}$ such that $\emptyset \neq \alpha \neq \mathbb{Q}$; if $p \in \alpha$ and $q < p$ then $q \in \alpha$; and α has no largest element. $\mathbb{R}$ is the set of cuts, ordered by inclusion.

Each rational r gives the cut $r^* = \{p \in \mathbb{Q} : p < r\}$, and $\{p \in \mathbb{Q} : p < 0\} \cup \{p \geq 0 : p^2 < 2\}$ is a cut not of this form.

Theorem A.1

(a) The order on $\mathbb{R}$ is total, and every nonempty set of cuts bounded above has a supremum, namely its union. (b) With $\alpha + \beta = \{p + q : p \in \alpha, q \in \beta\}$ and, for positive cuts, $\alpha\beta = \{pq : p \in \alpha, q \in \beta, p, q > 0\} \cup \{p \leq 0\}$ (extended to other signs by the usual rules), $\mathbb{R}$ is an ordered field. (c) $r \mapsto r^*$ embeds $\mathbb{Q}$ into $\mathbb{R}$ as an ordered subfield.

Proof. (a) If $\alpha \not\subset \beta$, pick $p \in \alpha \setminus \beta$; every $q \in \beta$ satisfies $q < p$ (otherwise $p \in \beta$), so $\beta \subset \alpha$. For A bounded by a cut β, the union $\gamma = \bigcup A$ is a cut ($\gamma \subset \beta \neq \mathbb{Q}$; downward closed; no maximum since no member has one), and it is clearly the least upper bound. (b) Most axioms are routine; the key points are the additive inverse $-\alpha = \{p : -p - r \notin \alpha$ for some rational $r > 0\}$ and the identity $\alpha + (-\alpha) = 0^*$, which uses the Archimedean property of $\mathbb{Q}$ to find $p \in \alpha$ with $p + w \notin \alpha$ for a given small $w > 0$. Multiplicative inverses are handled similarly for positive cuts. (c) The map preserves sums, products and order. ∎

Thus a complete ordered field exists, and by the uniqueness result of Chapter 1 it is unique up to isomorphism. Other constructions (equivalence classes of Cauchy sequences of rationals, or

infinite decimals) lead to the same field. The complex numbers are then $\mathbb{R}^2$ with $(a, b)(c, d) = (ac - bd, ad + bc)$.